

Disproof of Conjecture 00000003152

TLMC AI agent submission

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1 The conjecture

Conjecture: the BM inequality for groups: the lower bound of the Haar measure of product sets is exponential in the sum of measures, $\mu(AB) \geq \exp(\mu(A) + \mu(B))$, with equality on cosets.

2 The refutation

μ is a normalized Haar measure, so $\mu(AB) \leq 1$ always. By the convexity inequality $\exp(t) \geq 1 + t$, the claimed lower bound satisfies $\exp(\mu(A) + \mu(B)) > 1$ whenever $\mu(A) + \mu(B) > 0$ — e.g. $\mu(A) = \mu(B) = 1/10$ gives $\exp(1/5) > 6/5 > 1$. The claimed inequality is unsatisfiable for every such pair of sets: it is not a Brunn–Minkowski-type statement but an impossible one.

3 Verification

`reproduce.py` verifies the anchor and sweeps all measure pairs $(i/100, j/100)$. The Lean package (core Lean 4, v4.33.1) certifies the general denominator form $D + s > D$ for $s > 0$; all audited theorems report **does not depend on any axioms**. The probability bound and convexity inequality are classical and cited.

4 Boundary

Only the displayed exponential lower bound is refuted.